



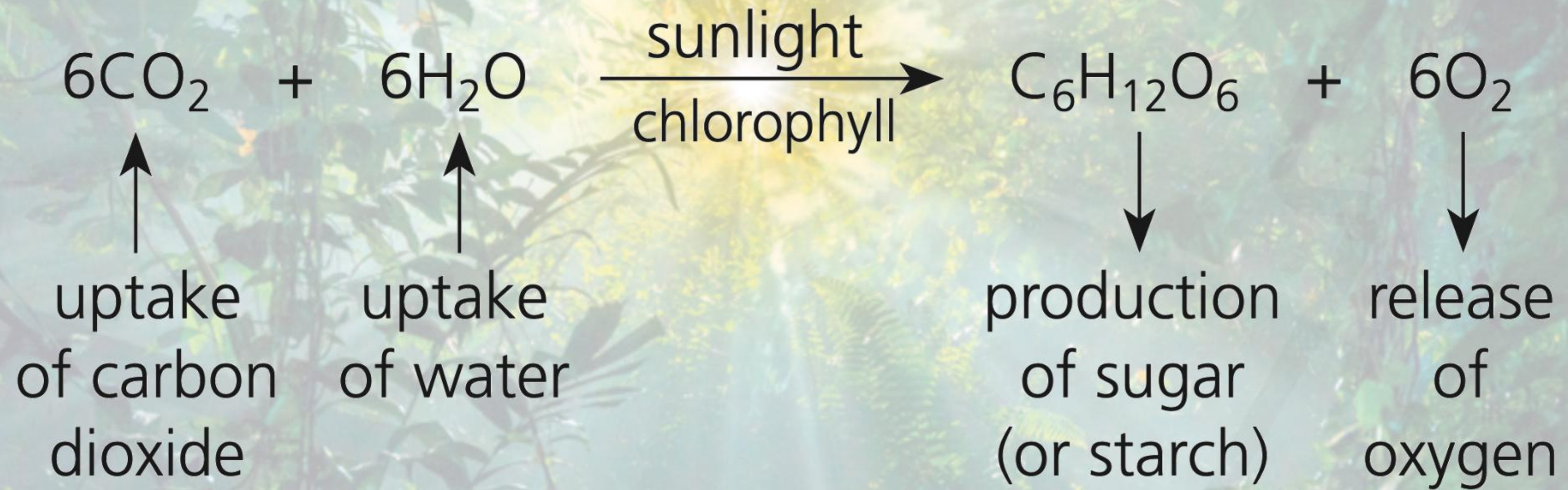
## Topic 7

# Photosynthesis

## Practical

**Unit 2** Biological Processes

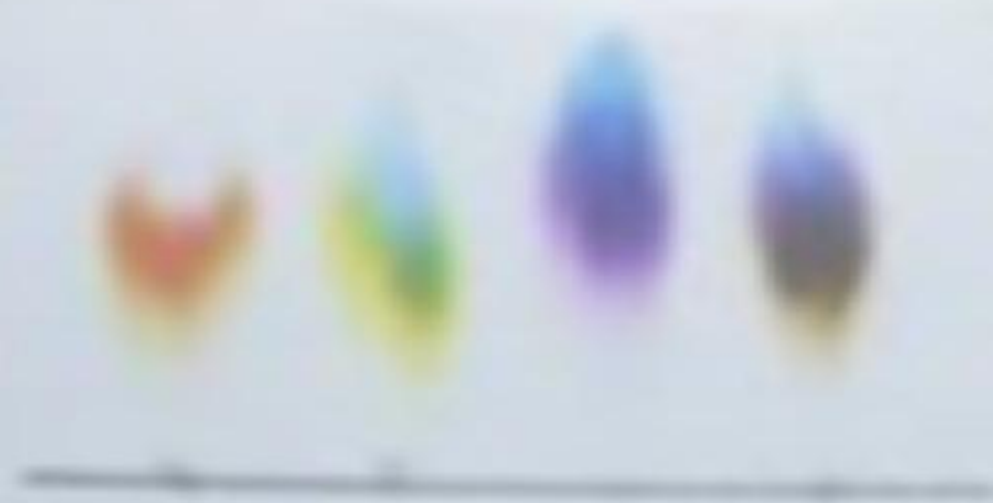
Bridge Program **BIOLOGY**



**Chlorophyll** 叶绿素 is a photosynthetic **pigment** that is found in **chloroplasts**.  
It **transfers energy** from light into energy in chemicals for the **synthesis** of carbohydrates.

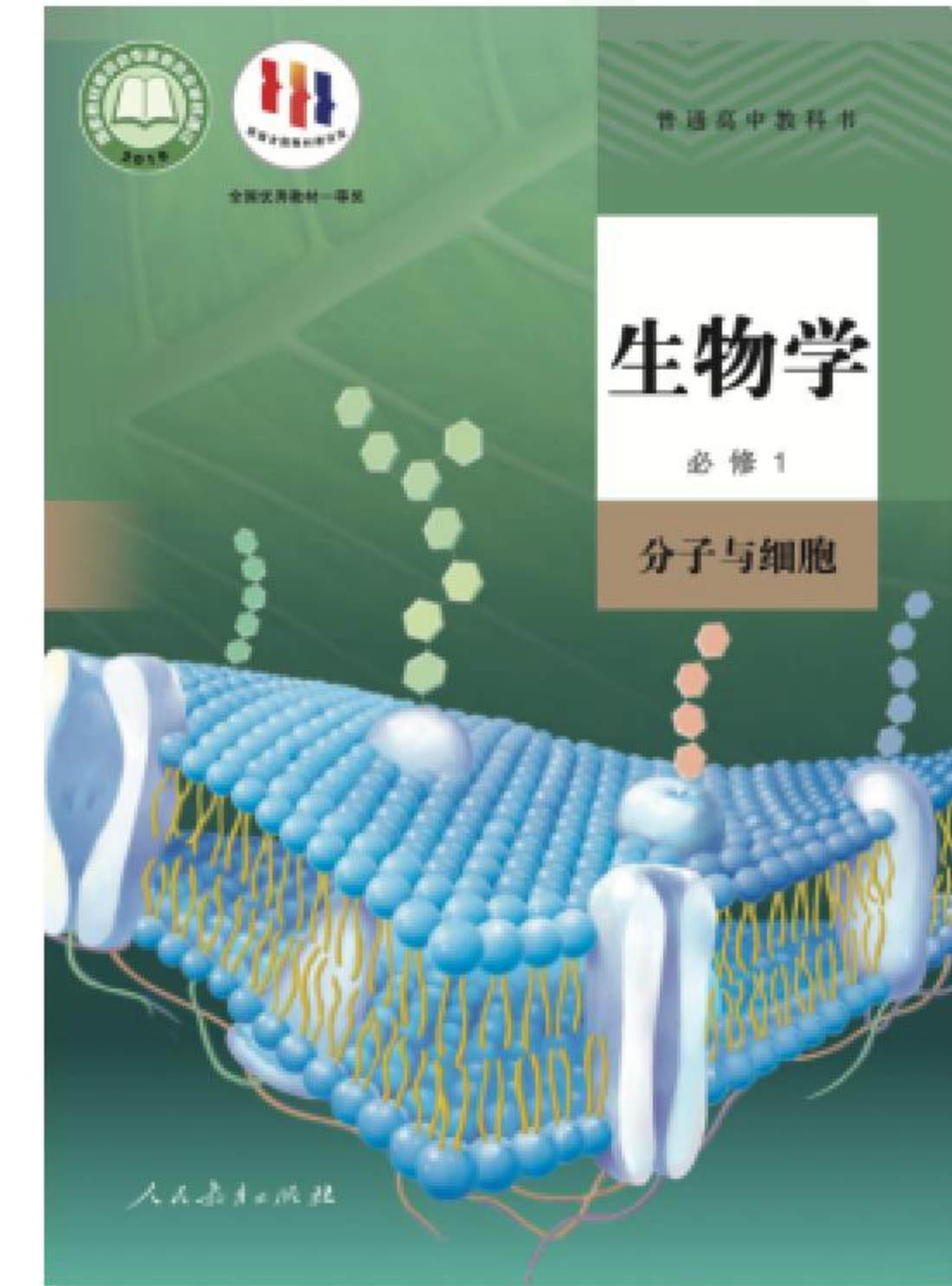
# Extraction and separation of photosynthetic pigments by paper chromatography

植物光合色素的提取和分离（纸层析法）



# LESSON OBJECTIVES

1. Extract **photosynthetic pigments** from plant leaves using a suitable solvent
2. Separate pigments using **paper chromatography**
3. **Identify chlorophyll a, chlorophyll b, carotene, and xanthophyll** based on color and position
4. **Calculate  $R_f$**  values for the four pigments
5. **Explain** how differences in **solubility** and **affinity** for the solvent affect pigment separation



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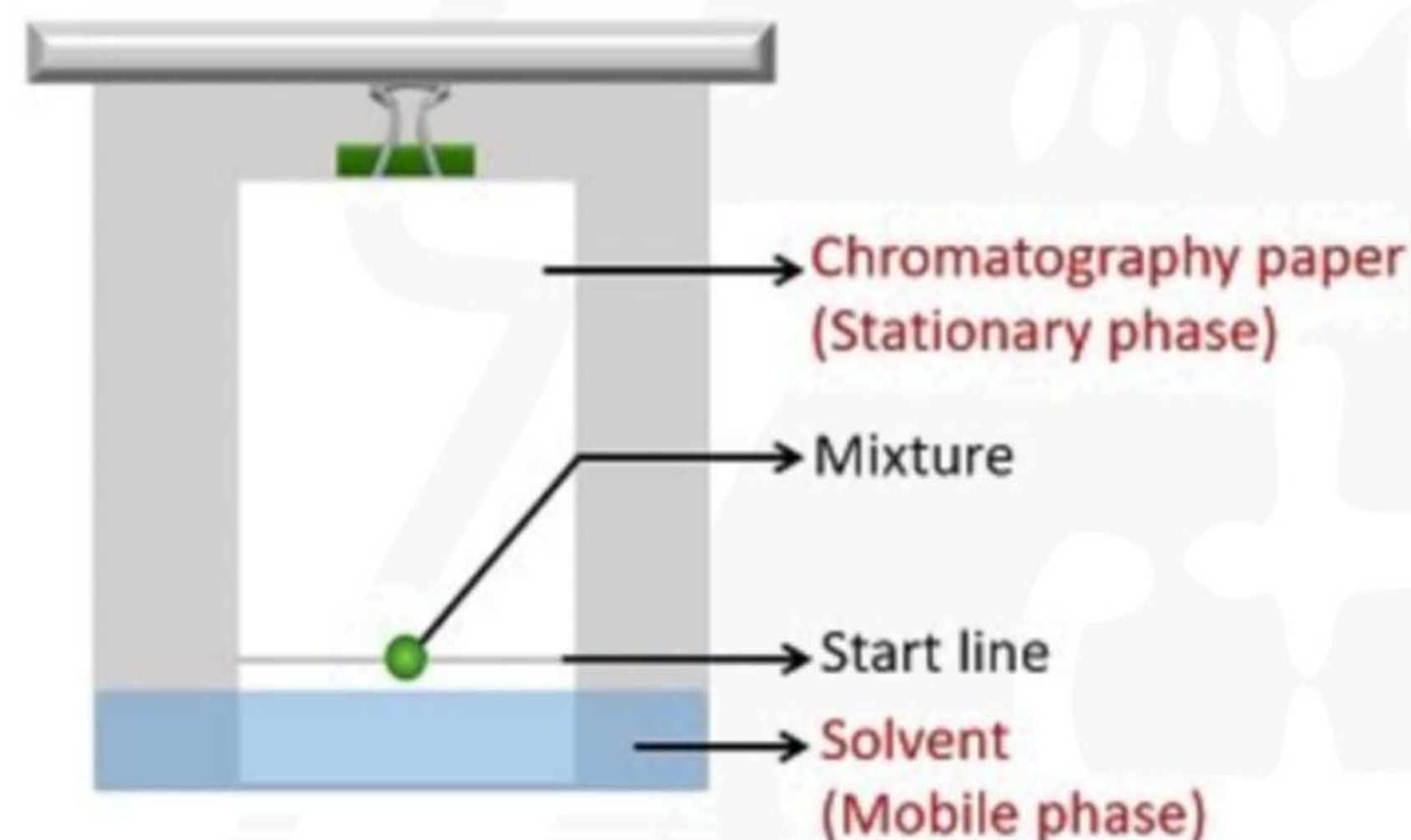
## PRINCIPLES

### Paper Chromatography 纸层析

It is the simplest chromatography method given by Christian Friedrich Schonbein in 1865. Paper chromatography uses filter paper with uniform porosity and high resolution.

The mixtures in compounds have **different solubilities**. For this reason, they get separated distinctly between the stationary and running phase.

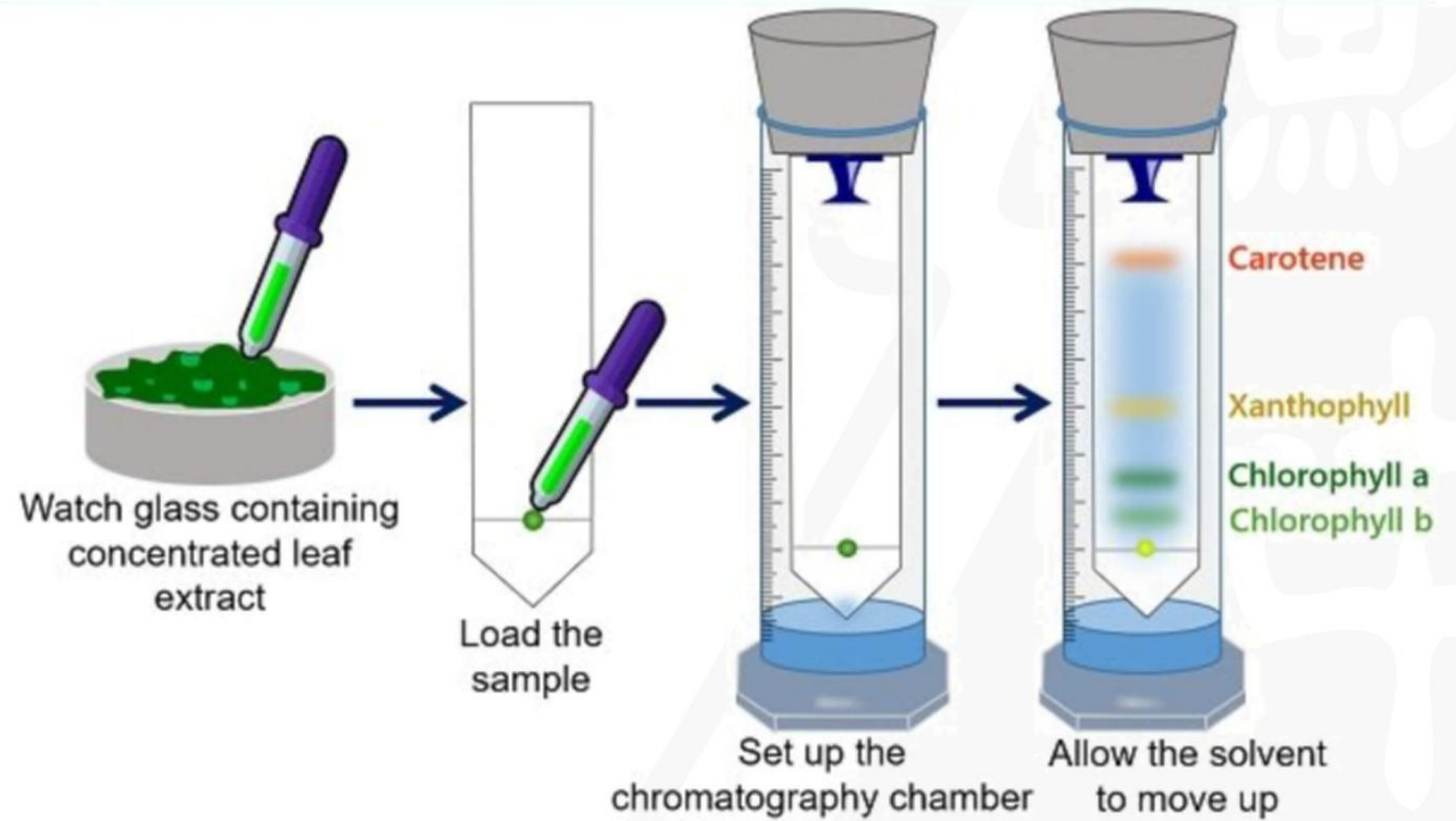
- The **mobile phase** is a combination of non-polar organic solvents. The solvent runs up the stationary phase via capillary movement.
- The **stationary phase** is polar inorganic solvent, i.e. water. Here, the absorbent paper supports the stationary phase, i.e. water.





## Lab procedures at a glance

- Prepare a concentrated leaf extract
- Apply a small sample spot on to the chromatography paper
- Place the paper into a chromatography chamber containing the solvent, ensuring the sample spot is above the solvent level
- Allow the solvent to travel up the paper to separate and visualize the colored pigments





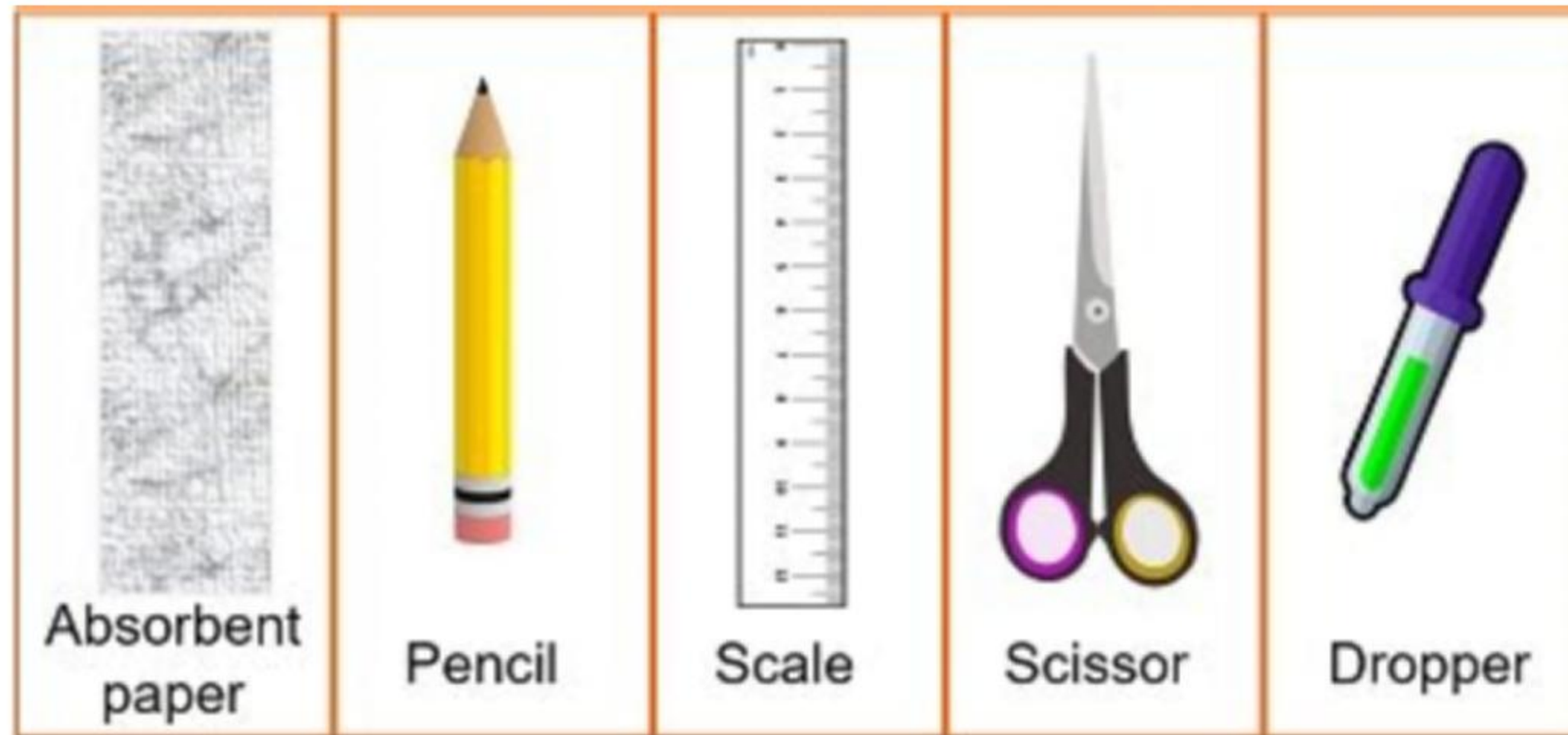
## A Prepare a concentrated leaf extract



- Weigh approximately **5 g** of spinach leaves
- Using scissors, cut the leaves into a mortar, avoiding the thick veins.
- Add approximately **5 mL** of anhydrous ethanol to the mortar
- Grind the spinach leaves with a pestle until a liquid paste is formed. Add a small amount of  $\text{SiO}_2$  to improve grinding. Add a small amount of  $\text{CaCO}_3$  to protect the pigments from degradation.
- Filter the mixture through a single layer of gauze into a **dry** test tube.
- Cork the test tube to prevent evaporation of the sample.



## B Apply the sample on to the chromatography paper

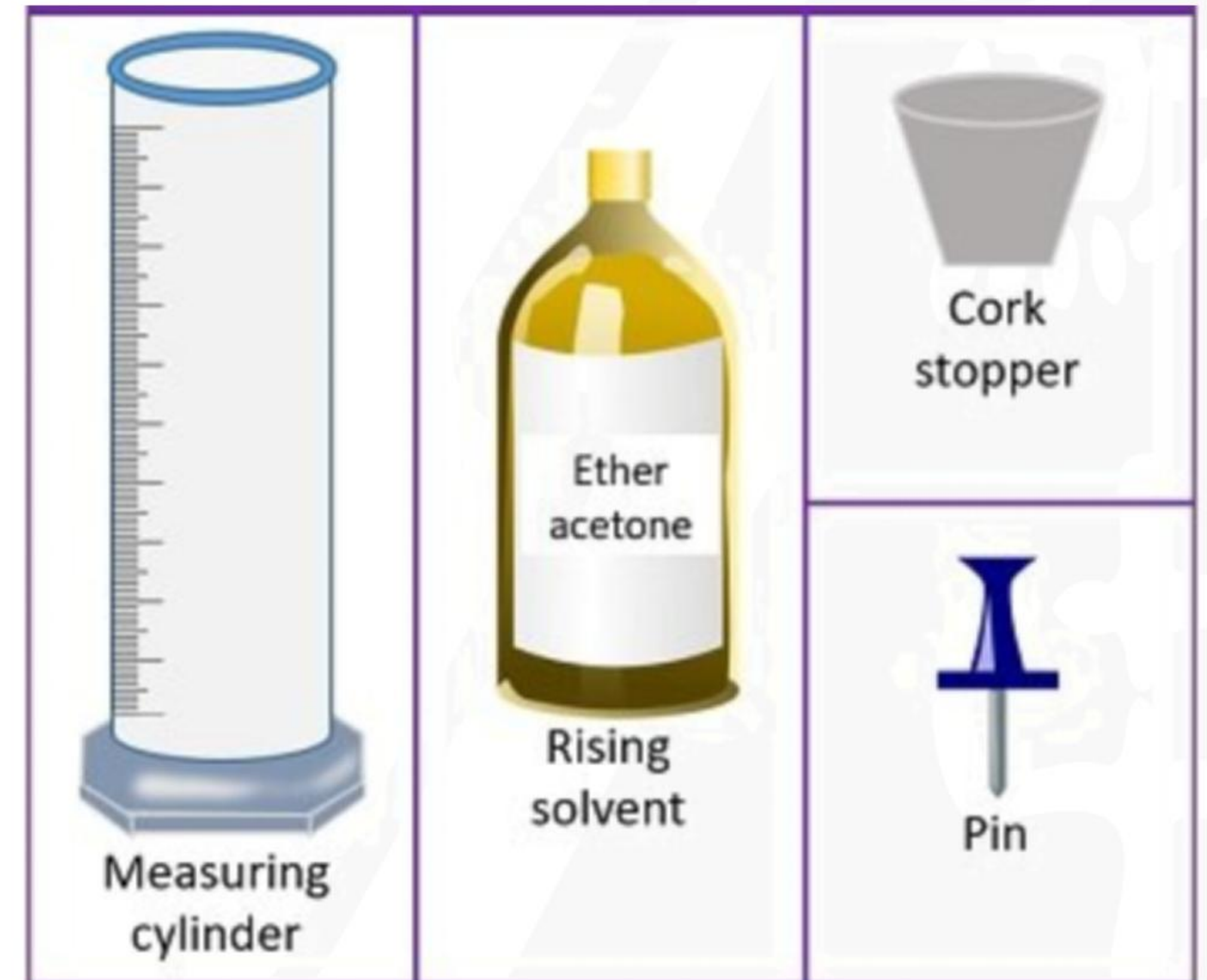


- Draw a faint horizontal line 1 cm from the bottom of the chromatography paper using a pencil and ruler.
- Cut the lower edge of the paper into a conical point below the pencil line.
- Using a capillary tube, place one small drop of leaf extract at the center of the pencil line. Allow the spot to dry completely.
- Repeat the above step 3–4 times to ensure the spot is sufficiently concentrated.



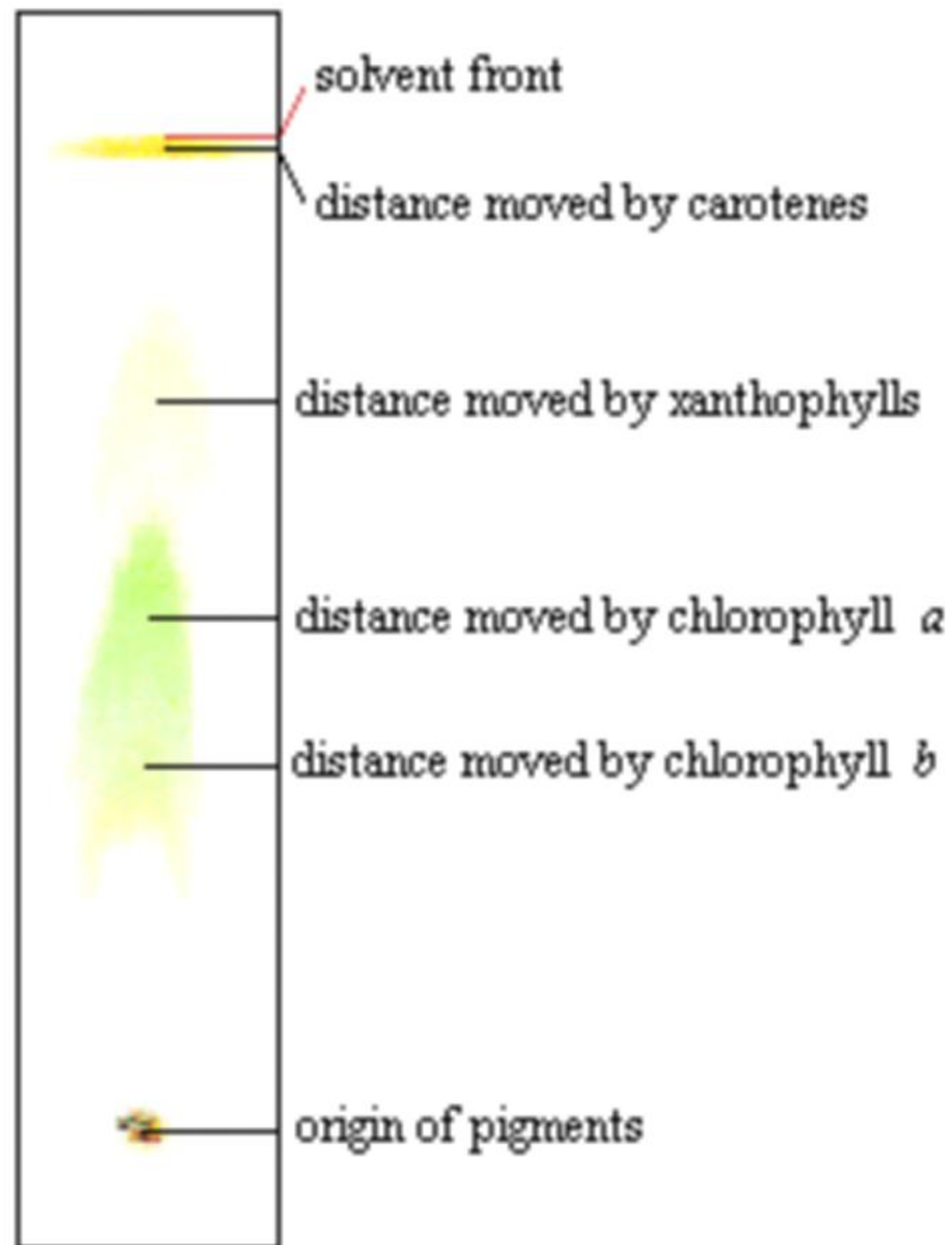
## C Run the chromatography

- Take a clean, dry chromatography chamber.
- Place the paper into the chamber, ensuring it does not touch the sides.
- Add the chromatography solvent to the chamber so that the solvent level is below the pigment spot.
- Cork the chamber immediately.
- Allow the solvent to rise up the paper by capillary action.
- When the solvent front stops moving or approaches the top, remove the paper.
- Allow the paper to dry completely.





## OBSERVATION



After drying, four distinct pigment bands should be visible:

- **carotene** 胡萝卜素: yellow-orange band at the top
- **xanthophyll** 叶黄素: yellow band below carotene
- **chlorophyll-a** 叶绿素a: dark green band
- **chlorophyll-b** 叶绿素b: light green band at the bottom

These bands form due to different **solubility** and **affinities** 亲和力 of pigments for the mobile phase (solvent) and the stationary phase (paper).

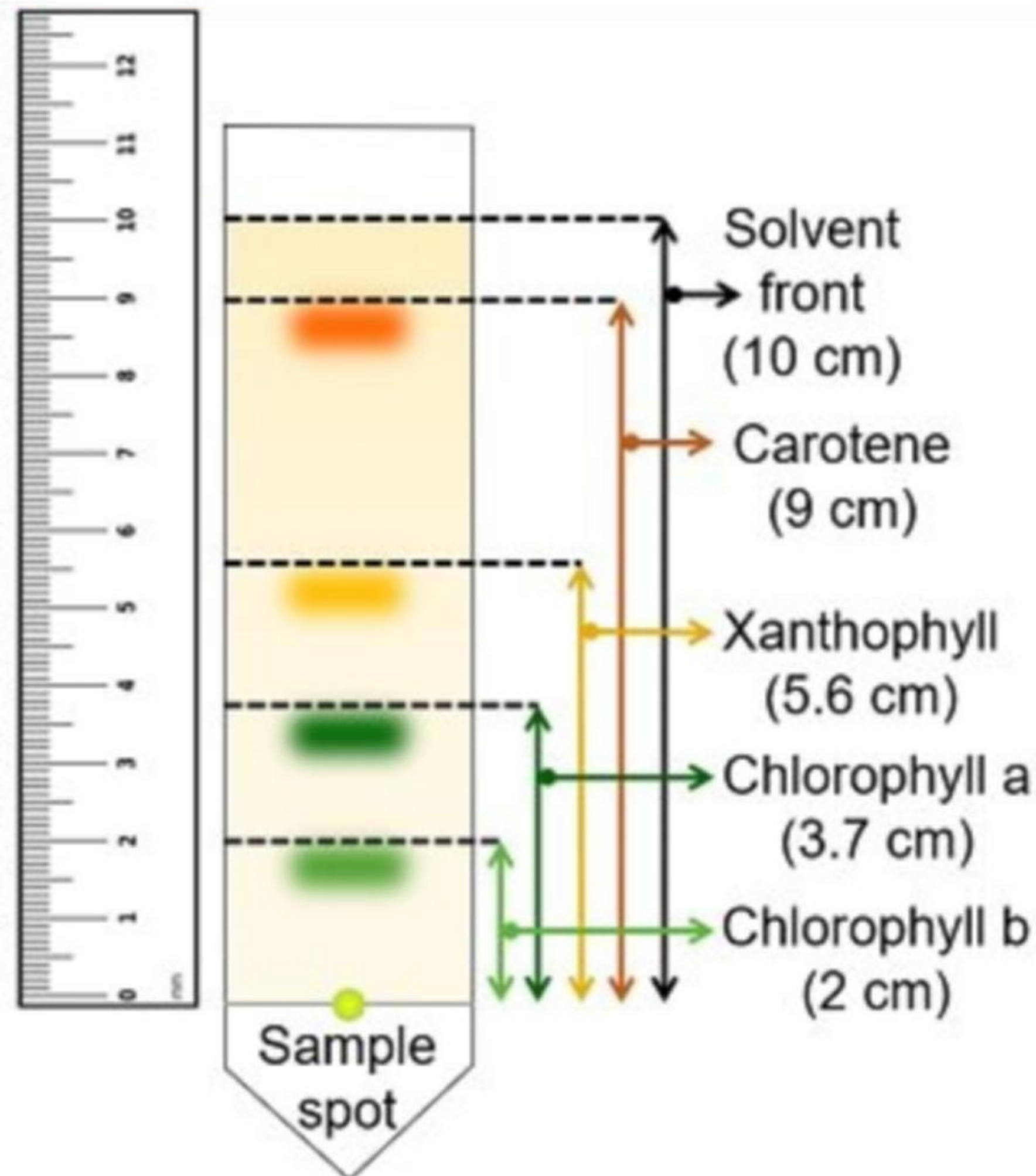


## Retention factor value ( $R_f$ ) 比移值

$$R_f = \frac{\text{distance moved by substance}}{\text{distance moved by solvent}}$$

$R_f$  values are used to identify the components of a mixture. The  $R_f$  value of a particular compound is **constant** when measured under the **same experimental conditions**. By calculating the  $R_f$  value, scientists can **identify unknown substances** by comparing them with the  $R_f$  values of **known substances** obtained using the same solvent and chromatography setup.

The  $R_f$  value is a **ratio** – it has **no units** and is always **less than 1**.





# Photosynthetic pigments 光合色素

Photosynthetic **pigments** are colored compounds that are present in the **thylakoid membrane** 类囊体膜 of the **chloroplast**. They **capture light energy**.

There are **two** types of pigments: **chlorophylls** 叶绿素 and **carotenoids** 类胡萝卜素. Pigments are colored because they absorb particular **wavelengths** of light and reflect others.

Chlorophyll is the pigment that gives plants green color by reflecting green light. Carotenoids reflect red, orange or yellow light.

